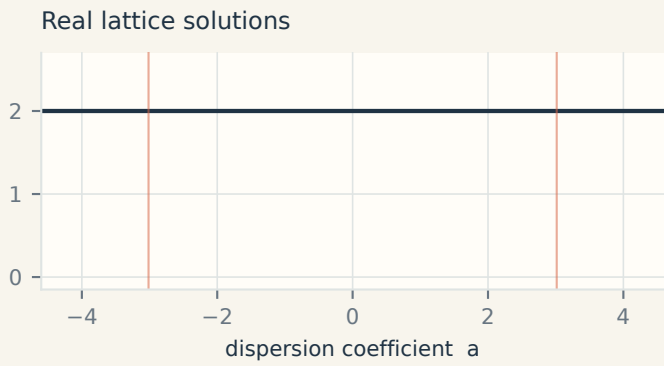
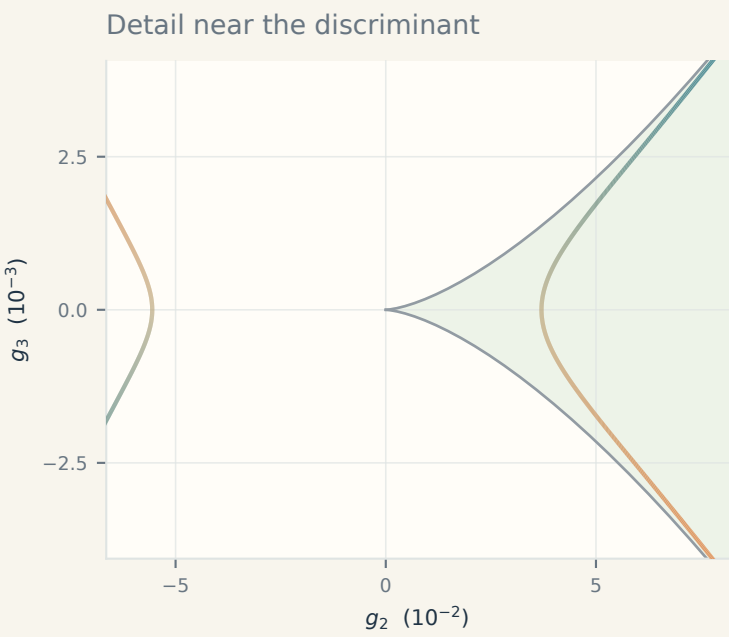
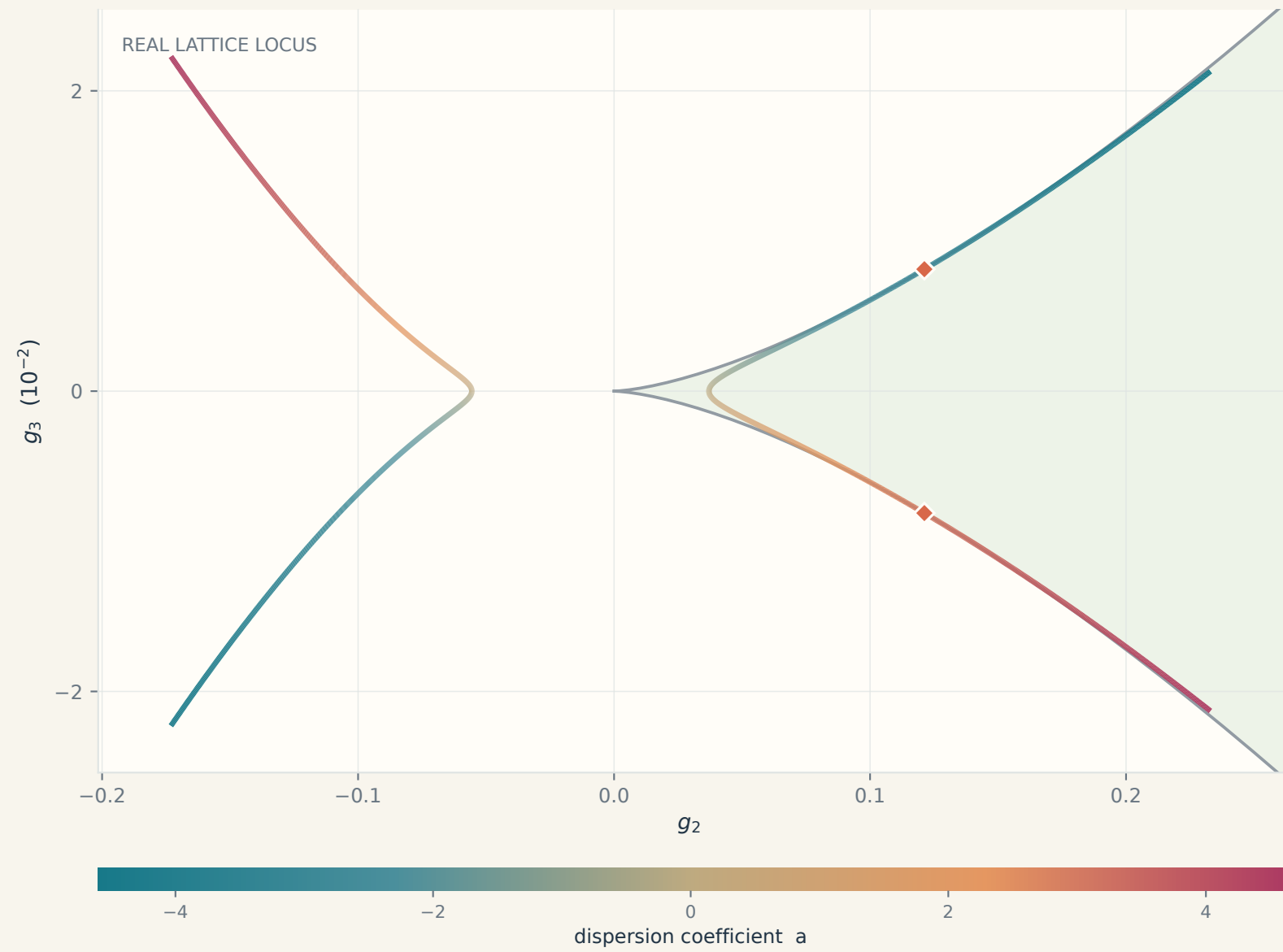


Cubic fourth order · two tangencies

$(n, p) = (3, 4) \cdot q = 2$

$u'''' + au'' - u - 120u^3 = 0$



Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Both character branches are shown, up to translation and sign.
The twisted branch touches the cusp at $a = \pm 10/\sqrt{11}$; the ordinary branch stays at $\Delta < 0$.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.